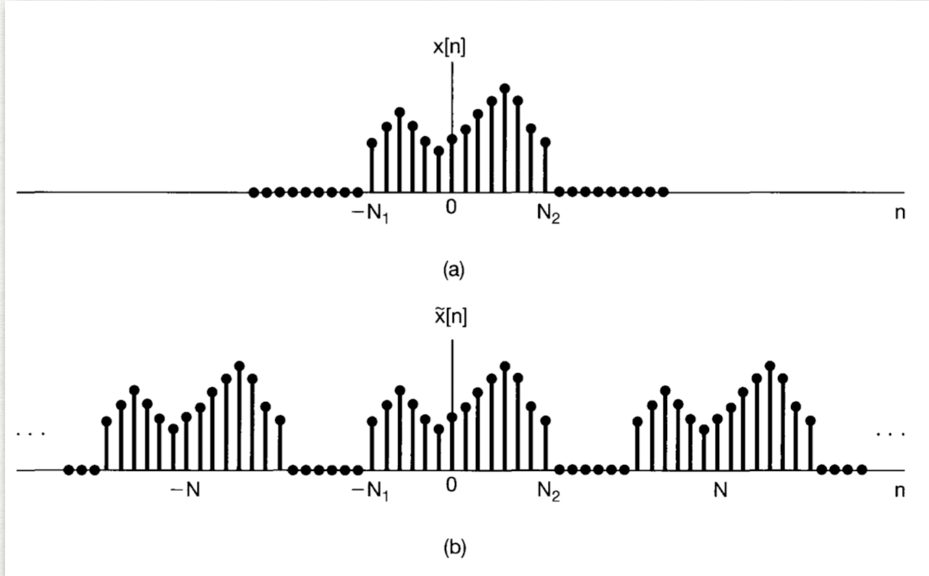
DISCRETE TIME FOURIER TRANSFORM

PROOF

- Fourier series representation of $\tilde{x}[n] = \sum_{k=<N>} a_k e^{jk(2\pi/N)n}$
- We apply analogous procedure to discrete-time signals. Considering x[n] of finite duration, $x[n] = 0$ when it is out of the range, $N_1 \leq n \leq N_2$. We construct a periodic signal $\tilde{x}[n]$. $x[n]$ is identical to $\tilde{x}[n]$ over a longer interval, that is $N \to \infty$, for any finite value of n.



(a) finite duration signal x[n]; (b) periodic signal $\tilde{x}[n]$ constructed to be equal to x[n] over one period.

DISCRETE TIME FOURIER TRANSFORM

PROOF

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$$\tilde{x}[n] = \sum_{k=<N>} a_k e^{jk(2\pi/N)n}$$
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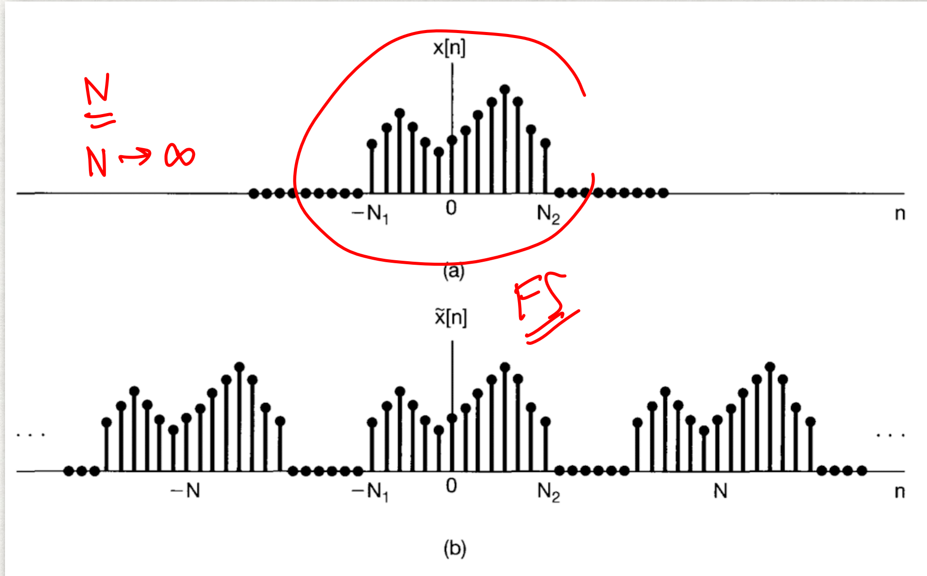


(a) finite duration signal x[n]; (b) periodic signal $\tilde{x}[n]$ constructed to be equal to x[n] over one period.

$N \to \infty$ $\quad \omega_0 = \frac{2\pi}{N} \to \frac{1}{N} \to \frac{\omega_0}{2\pi}$

$X(\Omega)$

$X(e^{j\omega})$

DISCRETE TIME FOURIER TRANSFORM

PROOF

it can be rewritten as

$$\tilde{x}[n] = \frac{1}{2\pi}\sum_{k=<N>} X(e^{jk\omega_0})e^{jk\omega_0 n}\omega_0 \qquad (4.12)$$

$k \to \omega = k\omega_0$

N increases ($N \to \infty$), ω_0 decreases , (4.12) passes to an integral.

$a_k \to N$

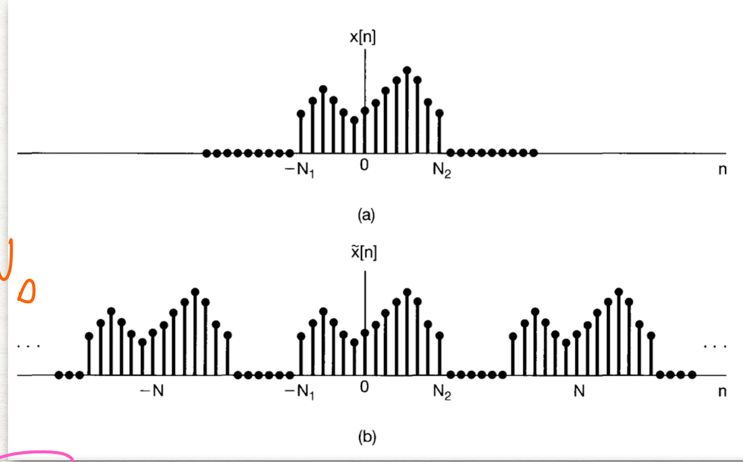
From (4.11), $X(e^{j\omega})$ is seen to be periodic.

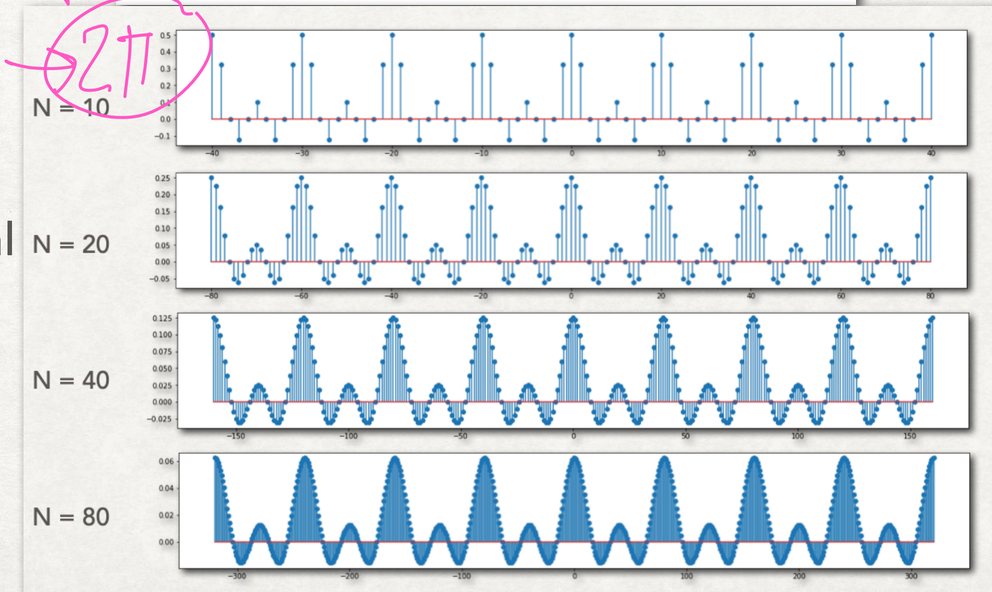
$X(e^{j\omega}) \to 2\pi$

As $\omega_0 \to 0$, sum become integral, sum carries out over N consecutive intervals of width $2\pi/N$. The total interval of integration will always have width of 2π. Therefore, $N \to \infty$ and $\tilde{x}[n] = x[n]$, (4.12) becomes

$$\boxed{x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega} \qquad (4.13)$$

ω





$k \to 0 \to N$

$\omega \to k\omega_0 \quad 0 \to N\omega_0 \quad \frac{2\pi}{N} \to 2\pi$

FT $X(e^{j\omega})$ DTFT

$\sum_{k=\langle N \rangle}$ k [0, N]   LF   0   π   HF   2π   w   LF

$\int$ $\omega = ?$ $[0, 2\pi]$ $\int_0^{2\pi}$

$\omega = k \cdot \omega_0 = \cancel{k} \cdot \frac{2\pi}{\cancel{N}}$

DISCRETE TIME FOURIER TRANSFORM

PROOF

Since $X(e^{j\omega})e^{j\omega n}$ is periodic with period 2π, the interval of integration can be taken as any interval of length 2π. We have the following pair of equations:

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}\, d\omega \qquad (4.14)$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \qquad (4.15)$$

(4.14) is a synthetic function; (4.15) is an analysis function

Also, FT $X(e^{j\omega})$ referred as spectrum of $x[n]$

CONVERGENCE ISSUES ASSOCIATED WITH THE DTFT

- $X(e^{j\omega})$ in (4.15) will converge either if $x[n]$ is absolutely summable, that is

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

- or if the sequence has finite energy, that is,

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty$$

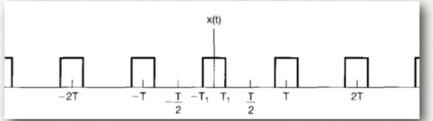
- (4.14) involves finite sum, so no associated convergence issues.

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}\,d\omega \qquad (4.14)$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \qquad (4.15)$$

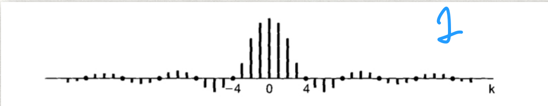
COMPARISON

FOURIER SERIES

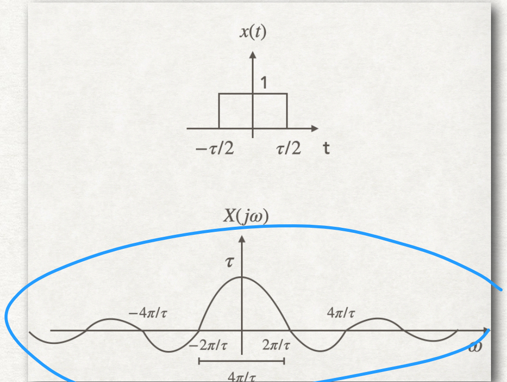


$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t}$$

$$a_k = \frac{1}{T}\int_{<T>} x(t)e^{-jk\omega_0 t}\,dt = \frac{1}{T}\int_0^T x(t)e^{-jk(2\pi/T)t}dt$$



C T FOURIER TRANSFORM

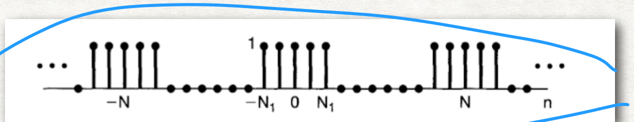


$$x(t) = \frac{1}{2\pi}\int_{\omega=-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega$$

$$X(j\omega) = \int_{t=-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$$

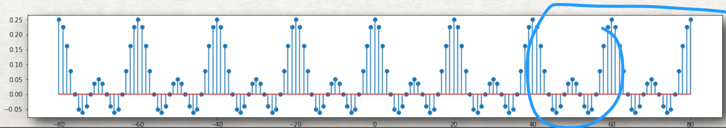
CONTINUOUS-TIME

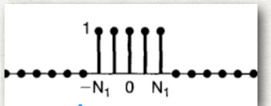
DISCRETE-TIME



$$x[n] = \sum_{k=<N>} a_k e^{jk(\omega_0)n} = \sum_{k=<N>} a_k e^{jk(2\pi/N)n}$$

$$a_k = \frac{1}{N}\sum_{n=<N>} x[n]e^{-jk\omega_0 n} = \frac{1}{N}\sum_{n=<N>} x[n]e^{-jk(2\pi/N)n}$$





?

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}\,d\omega$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

DISCRETE-TIME FOURIER TRANSFORM

EXERCISE #3

Considering a signal $x[n] = a^n u[n]$ where $|a| < 1$

From $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$

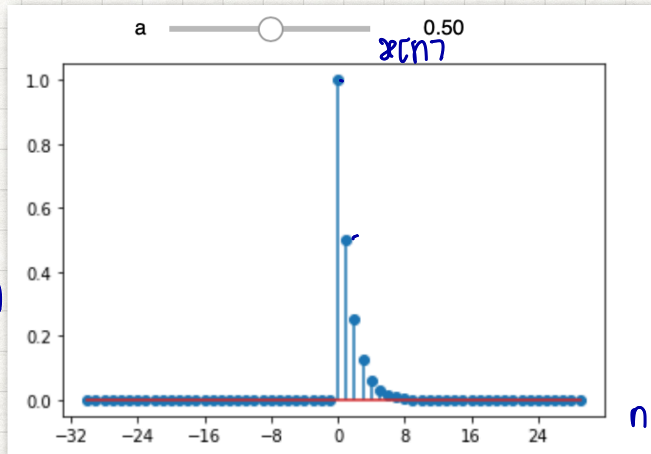
$= \sum_{n=0}^{\infty} a^n e^{-j\omega n}$

$= \frac{1}{1 - ae^{-j\omega}}$

$r \rightarrow ae^{-j\omega}$

$|X(e^{j\omega})|$, $\measuredangle X(e^{j\omega})$

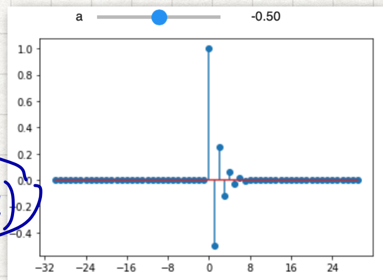
Re, Im

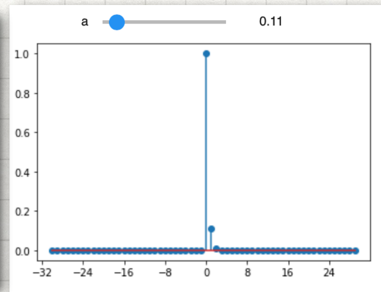


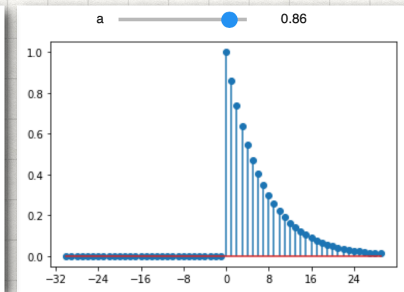
$\omega = 0$ $\quad X(e^{j\omega}) = \frac{1}{1-a}$ $\quad \measuredangle X(e^{j\omega}) = 0$

$\omega = \frac{\pi}{2}$ $\quad X(e^{j\omega}) = \frac{1}{1-ae^{-j\frac{\pi}{2}}}$ $\quad \left|\frac{1}{1+aj}\right| = \frac{1}{\sqrt{1+a^2}}$ $\quad \measuredangle = \tan^{-1}(-a)$

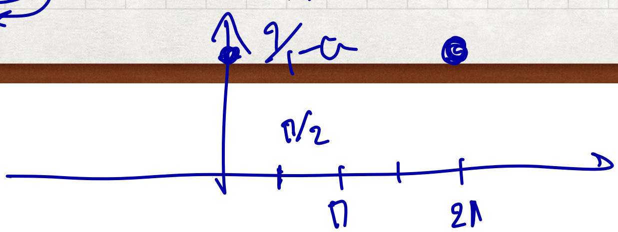
$\omega = \pi$ $\quad \frac{1}{1-ae^{-j\pi}} = \frac{1}{1+a}$







$|X(e^{j\omega})|$



DISCRETE-TIME FOURIER TRANSFORM

EXERCISE #3

Considering a signal $x[n] = a^n u[n] \quad where\,|a| < 1$

From $X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} x[n]e^{-j\omega n}$, we will have

$$X(e^{j\omega}) = \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \sum_{n=0}^{\infty} (ae^{-j\omega})^n$$

$$= \frac{1}{1 - ae^{-j\omega}}$$

$$\boxed{a^n u[n] \leftrightarrow \frac{1}{1 - ae^{-j\omega}}}$$

fnte

$X(e^{j\omega})$

$a \uparrow$

$\downarrow$

$\ominus$

Geometric Series

$$a + ar + ar^2 + \ldots + ar^N = \sum_{k=0}^{N} ar^k = a\left(\frac{1 - r^{N+1}}{1 - r}\right)$$

$$a + ar + ar^2 + \ldots = \sum_{k=0}^{\infty} ar^k = a\left(\frac{1}{1 - r}\right),\text{ for |r|<1}$$

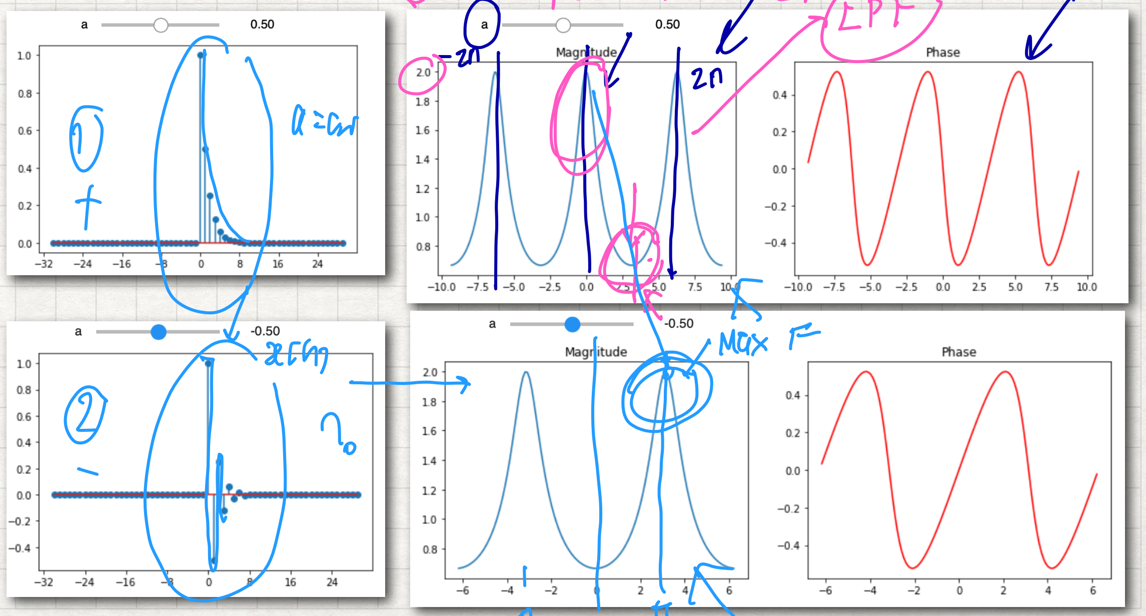
$a < 0.5$

$|X(e^{j\omega})|$

HF

LF

LPF



HPF

DISCRETE-TIME FOURIER TRANSFORM

EXERCISE #3

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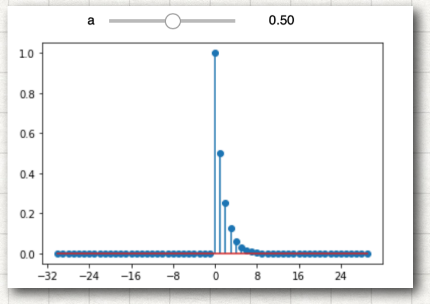
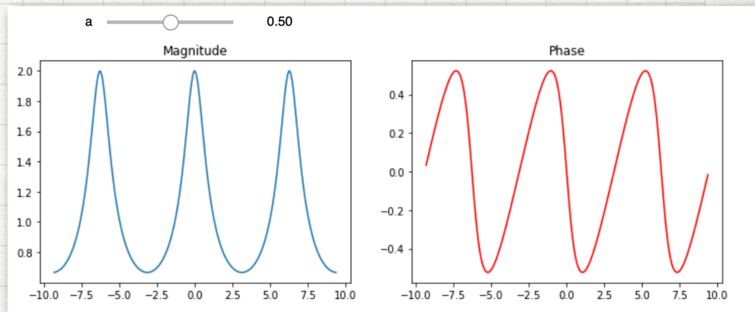
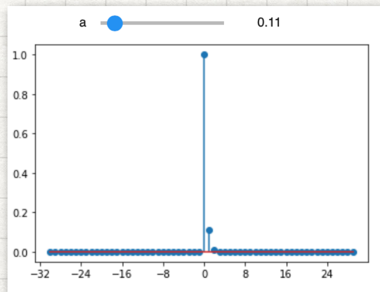
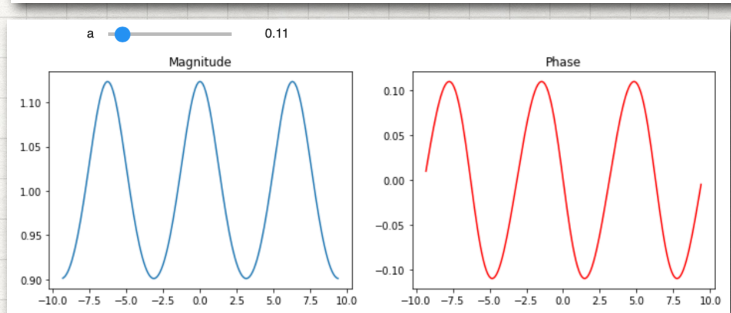
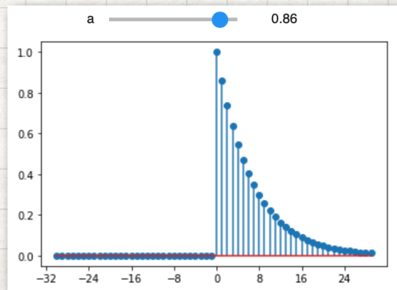
$$= \frac{1}{1 - ae^{-j\omega}}$$

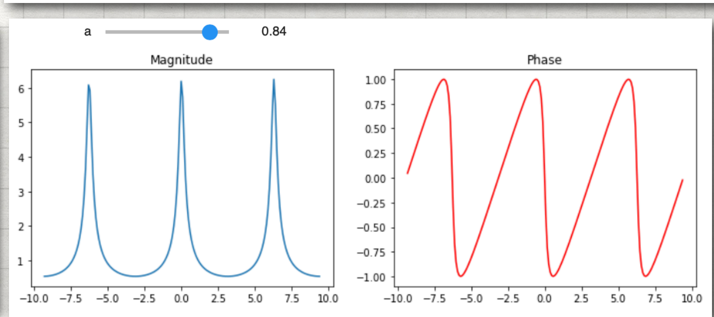






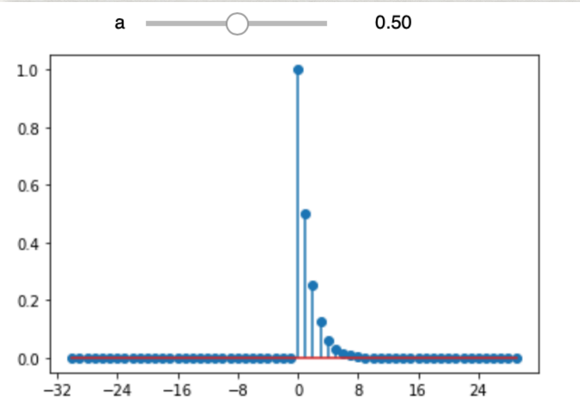


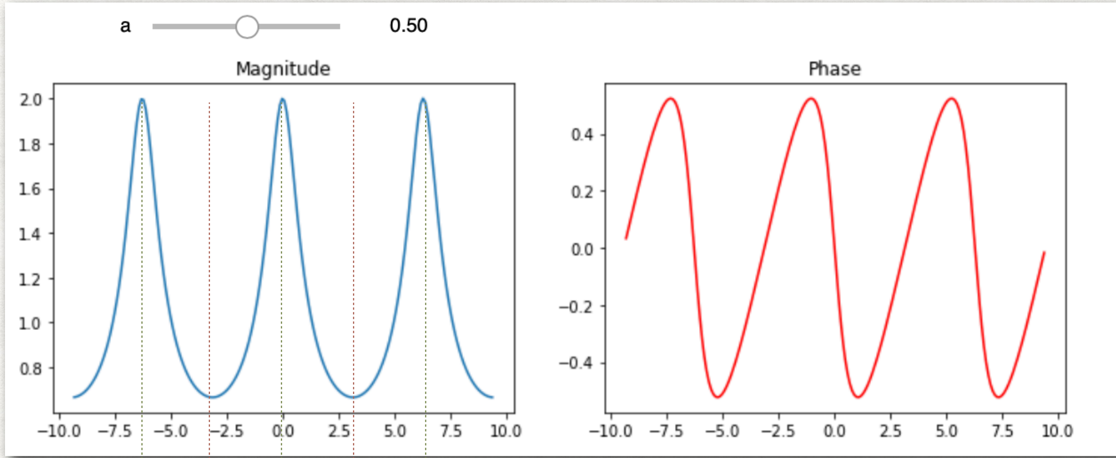




DISCRETE-TIME FOURIER TRANSFORM

EXERCISE #3





-2π $-\pi$ 0 π 2π

LF

$X(e^{j\omega})$ is periodic with period 2π; $\omega = 0$ and $\omega = 2\pi$ yield the same signal. Signals at frequencies near these values or any other even multiple of π are thought of as low-frequency signals. Similarly, the high frequencies in discrete-time are ω near odd multiple of π.

$\boxed{1\pi, \quad 3\pi, \quad 5\pi}$

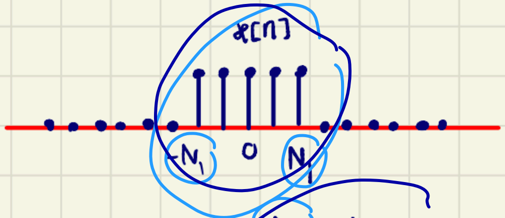
14:42

Example #4

Consider the rectangle pulse

$$x[n] = \begin{cases} 1, & |n| \le N_1 \\ 0, & |n| > N_1 \end{cases}$$

Giving $N_1 = 2$, $x[n]$ is shown below.



We get $X(e^{j\omega}) = \sum_{n=-N_1}^{N_1} e^{-j\omega n}$

$m = n + N_1$

$$= \sum_{m=0}^{2N_1} e^{-j\omega(m-N_1)}$$

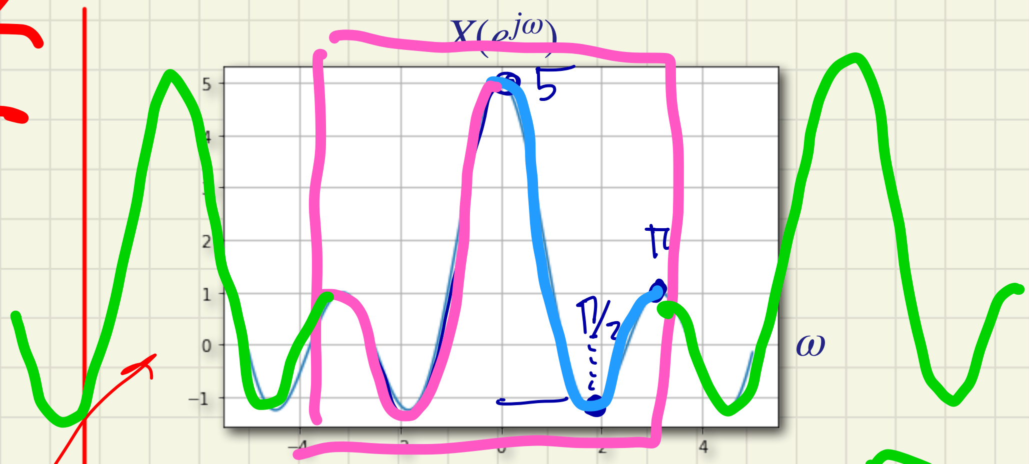
$$\sum_{k=r}^{N} ar^k = \frac{a(1-r^{N+1})}{1-r}$$

$$= e^{j\omega N_1}\left(\frac{1-e^{-j\omega(2N_1+1)}}{1-e^{-j\omega}}\right)$$

หา $e^{j\omega N_1}$ คูณกระจาย และ ดึง $e^{-j\omega(\frac{1}{2})}$ ออกมา

$$= \frac{\cancel{e^{-j\omega/2}}}{\cancel{e^{-j\omega/2}}}\left(\frac{e^{j\omega(N_1+\frac{1}{2})} - e^{-j\omega(N_1+\frac{1}{2})}}{e^{j\omega/2} - e^{-j\omega/2}}\right)$$

$$= \frac{\sin(\omega(N_1+\frac{1}{2}))}{\sin(\omega/2)}$$



$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$

$X(e^{j\omega})$

$$\frac{e^{-j\frac{\omega}{2}}\left(e^{j\omega(N_1+\frac{1}{2})} - e^{-j\omega(N_1+\frac{1}{2})}\right)}{e^{-j\frac{\omega}{2}}\left(e^{+j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}}\right)}$$

$N_1 = 2$

2π

$\omega = 0 \quad = 2N_1 + 1 = 5$

$\omega = \frac{\pi}{2} \quad = \frac{\sin(\frac{\pi}{2}(\frac{5}{2}))}{\sin(\pi/4)} = -1$

$\omega = \pi \qquad \frac{\sin(\pi(5/2))}{\sin(\pi/2)} = 1$

$* = \omega = -\pi$

2π

Example #5

Let $x[n]$ be the unit impulse, that is,

$$x[n] = \delta[n]$$

$\delta(t) \leftrightarrow 1$ $X(j\omega)$

We get

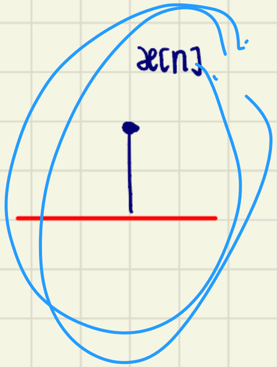
$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \cancel{x[n]}^{\,\delta[n]} e^{-j\omega n}$$

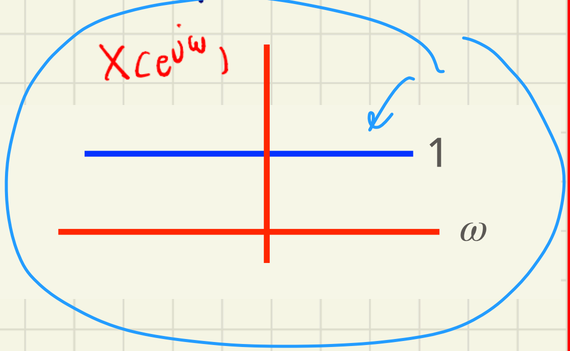
sampling → sifting

$$= \sum_{n=-\infty}^{\infty} 1 \cdot \delta[n]$$

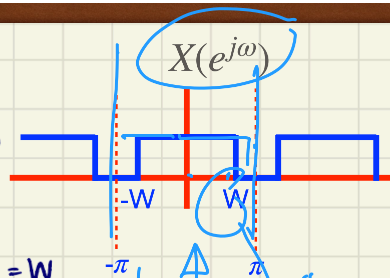
$$= 1$$

The unit impulse has a Fourier transform representation consisting of equal contributions at all frequencies.





$$\tilde{x}[n] = \frac{1}{2\pi}\int_{-W}^{W} e^{j\omega n}\, d\omega$$



$$= \frac{1}{2\pi}\left.\frac{e^{j\omega n}}{jn}\right|_{\omega=-W}^{\omega=W}$$

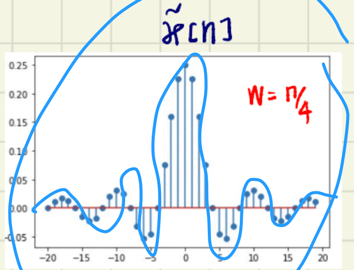
$$= \frac{1}{2\pi}\left(\frac{e^{jWn}}{jn} - \frac{e^{-jWn}}{jn}\right)$$

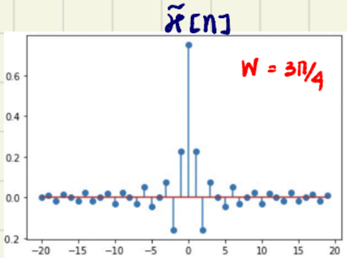
$$= \frac{1}{\cancel{2}\pi}\left(\frac{\cancel{2}\sin Wn}{n}\right)$$

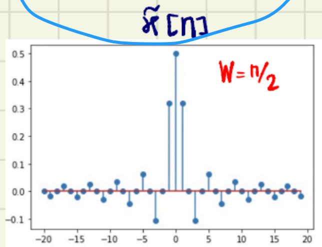
$$= \frac{\sin Wn}{\pi n}$$

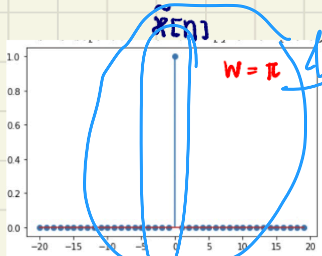
π

W









The Fourier transform for periodic signals

As in continuous-time case, discrete-time periodic signals can be incorporated within the framework of the discrete-time Fourier transform by interpreting the transform of a periodic signal as an impulse train in the frequency domain. To derive the form of this representation, consider the signal

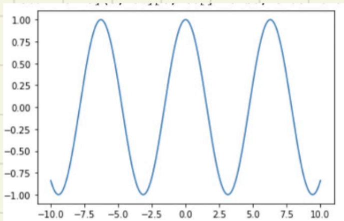
$$x[n] = e^{j\omega_0 n}$$

$X(j\omega)$
$\delta(\omega-\omega_0)$

In continuous-time, $x(t) = e^{j\omega_0 t} \Longleftrightarrow 2\pi\,\delta(\omega-\omega_0)$

e^{j1t}

$\omega_0 = 1$

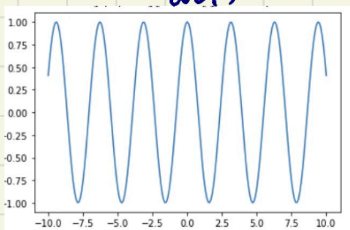


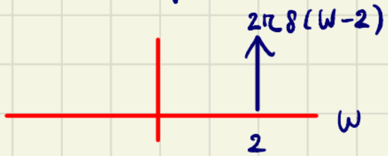
$x(t)$

$X(j\omega)$

e^{j2t}

$\omega_0 = 2$



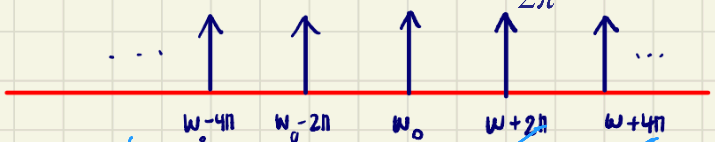


In discrete-time, we might expect for the same transform to result for the discrete-time signal. However, DTFT must be periodic in ω with period 2π.

The Fourier transform of $x[n]$ should have impulses at ω_0, $\omega_0+2\pi$, $\omega_0+4\pi$, ...

$$X(e^{j\omega}) = \sum_{l=-\infty}^{\infty} 2\pi\,\delta(\omega-\omega_0-2\pi l)$$

$X(e^{j\omega})$



$X(e^{j\omega})$ $\qquad e^{j\omega_0 n} \Leftrightarrow \sum_{l=-\infty}^{\infty} \delta(\omega-\omega_0-2\pi l)$

check the validity by taking inverse DTFT,

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega = \cancel{\frac{1}{2\pi}}\int_{2\pi} \sum_{l=-\infty}^{\infty} \cancel{2\pi}\,\delta(\omega-\omega_0-2\pi l)\, e^{j\omega n}\, d\omega$$

$$= \cancel{e^{j(\omega_0+2\pi r)n}}$$

$$= e^{j\omega_0 n}$$

aperiodic → FT only

Periodic → FS, FT

Now considering a periodic sequence

$$x[n] = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/N)n}$$

$$X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} x[n]e^{-j\omega n}$$

we get

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \overbrace{\sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/N)n}}^{x[n]} e^{-j\omega n}$$

SWAP

$$= \sum_{k=-\infty}^{\infty} a_k \sum_{n=-\infty}^{\infty} e^{-j(\omega - k\cdot\frac{2\pi}{N})n}$$

$$= \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - \frac{2\pi k}{N})$$

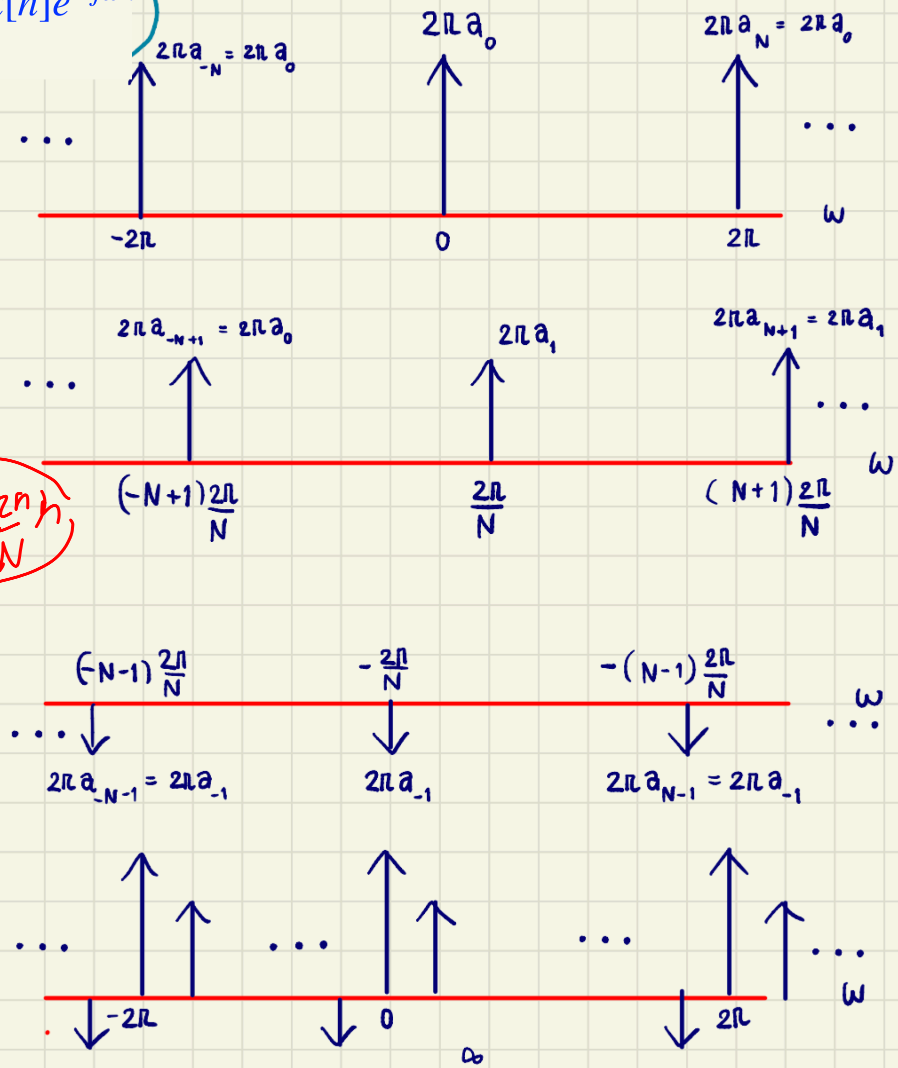
$\neq e^{-j(\omega - k\frac{2\pi}{N})n}$

Similar to continuous-time Fourier transform, Fourier transform of a discrete-time periodic signal can be directly constructed from its Fourier series coefficients.

per

① a_k

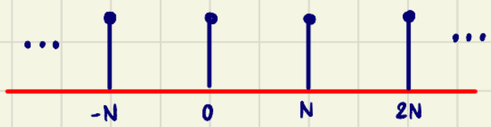
②



Fourie transform $X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - \frac{2\pi}{N}k)$

The discrete-time counterpart of the periodic impulse train in the sequence

$$x[n] = \sum_{k=-\infty}^{\infty} \delta[n-kN]$$



peri

Fourier series coefficients can be calculated by

$$a_k = \frac{1}{N} \sum_{n=\langle N\rangle} x[n]\, e^{-jk(2\pi/N)n}$$

Choosing the interval of summation as $0 \le n \le N-1$, we have

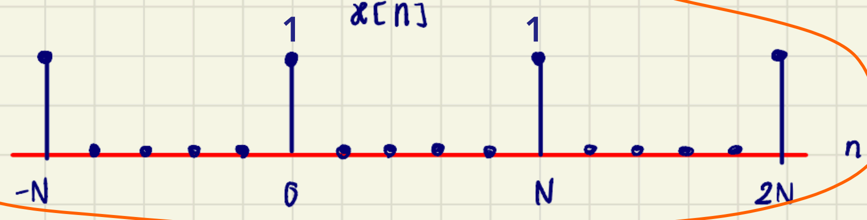
$$a_k = \frac{1}{N} \sum_{n=\langle N\rangle} \underbrace{\delta[n]\, e^{-jk\left(\frac{2\pi}{N}\right)n}}_{1}$$

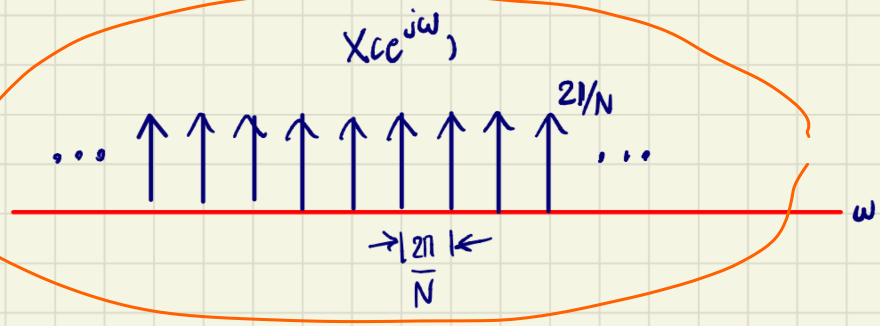
$$= \frac{1}{N}$$

$$x[n] = \sum_{k=\langle N\rangle} \frac{1}{N}\, e^{jk(2\pi/N)n}$$

$$X(e^{j\omega}) = \text{DTFT}\left\{\sum_{k=\langle N\rangle} \frac{1}{N} \cdot e^{jk\left(\frac{2\pi}{N}\right)n}\right\} = \underbrace{\sum_{k=\langle N\rangle} \frac{1}{N} \cdot 2\pi\, \delta\left(\omega - \frac{2\pi k}{N}\right)}_{0 \le n \le N-1}$$

$-2\pi l$





PROPERTIES OF THE DISCRETE TIME FOURIER TRANSFORM

Notation: $x[n] \overset{DTFT}{\leftrightarrow} X(e^{j\omega})$

$e^{j(\omega+2\pi)\cdot n}$ $e^{j\omega n}$

1. Periodicity $\quad X(e^{j\omega}) = X(e^{j(\omega+2\pi)})$

2π

2. Linearity of the FT

$$a x_1[n] + b x_2[n] \leftrightarrow a X_1(e^{j\omega}) + b X_1(e^{j\omega})$$

3. Time & Frequency Shifting

CTFT $\quad t_0$

(Time shifting) $\quad x[n - n_0] \leftrightarrow e^{-j\omega n_0} X(e^{j\omega})$

$n-n_0$

(Frequency shifting) $\quad e^{j\omega_0 n} x[n] \leftrightarrow X(e^{j(\omega-\omega_0)})$

$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$

$x[n-n_0]$ $\quad X(e^{j\omega}) e^{-j\omega n_0}$

$x[n]$ $\quad \hat{x}(\omega)$

4. Conjugation and Conjugate Symmetry

$\omega \to -\omega$

If $x[n]$ is real $\rightarrow$ $\boxed{X(e^{-j\omega}) = X^*(e^{j\omega})}$

$$even \begin{cases} Re\{X(e^{j\omega})\} \\ |X(e^{j\omega})| \end{cases} \qquad odd \begin{cases} Im\{X(e^{j\omega})\} \\ \measuredangle X(e^{j\omega}) \end{cases}$$

5. Differencing and Accumulation

$X(e^{j\omega}) - e^{-j\omega} X(e^{j\omega})$

Differencing: $\quad x[n] - x[n-1] \leftrightarrow (1 - e^{j\omega})X(e^{j\omega})$

Accumulation: $\quad \ldots X(e^{j\omega})$

Considering $y[n] = \sum_{m=-\infty}^{n} x[m]$

$$y[n] \leftrightarrow \frac{X(e^{j\omega})}{1 - e^{-j\omega}} + \pi X(e^{j0}) \sum_{l=-\infty}^{\infty} \delta(\omega - 2\pi l)$$

$y[n] = x[n] * u[n]$

$x[n] * h[n] \leftrightarrow X(e^{j\omega}) \cdot H(e^{j\omega})$

PROPERTIES OF THE DISCRETE TIME FOURIER TRANSFORM

$u[n] \leftrightarrow U(e^{j\omega})$ *

$\omega = 0$

$u[n] - u[n-1] = \delta[n]$

$U(e^{j\omega}) - e^{-j\omega}U(e^{j\omega}) = 1$

$U(e^{j\omega}) = \frac{1}{1-e^{-j\omega}} + C\,\delta(\omega)$

$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$

$\sum_{l=-\infty}^{\infty} \delta(\omega - 2\pi l)$

Notation:

6. Time Expansion

$$x_{(k)}[n] = \begin{cases} x[n/k], & if\ n = multiple\ of\ k \\ 0, & if\ n \neq multiple\ of\ k \end{cases} \leftrightarrow X(e^{jk\omega})$$

7. Differentiation in Frequency

$$n\,x[n] \leftrightarrow j\frac{dX(e^{j\omega})}{d\omega}$$

$x[n]$

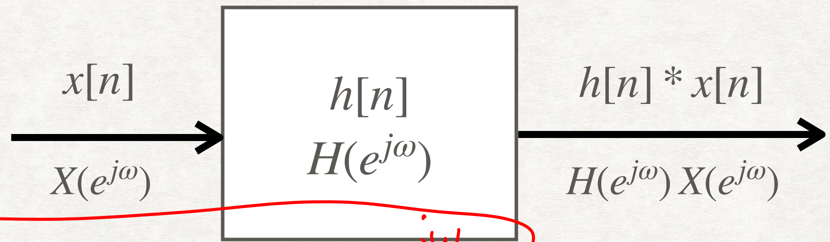
$\frac{dX(e^{j\omega})}{d\omega}$

8. Parseval's relation

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi}\int_{2\pi} |X(e^{j\omega})|^2 d\omega$$

9. Convolution properties

$$h[n] * x[n] \leftrightarrow H(e^{j\omega})X(e^{j\omega})$$



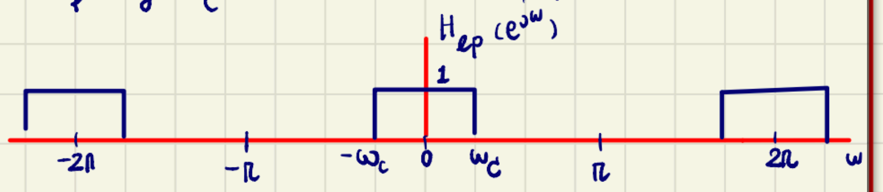
$e^{-j\omega n}$

$(-jn)\,x[n] \leftrightarrow \frac{dX(e^{j\omega})}{d\omega}$

$u[n]$, $u(t)$

Example of frequency-shifting

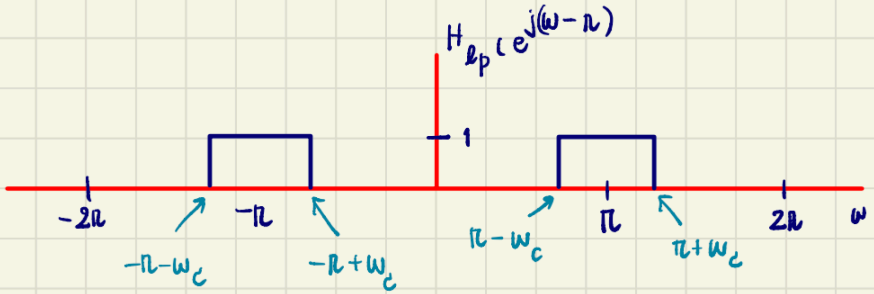
Frequency response $H_{lp}(e^{j\omega})$ of a lowpass filter with cutoff frequency ω_c.



Shift $H_{lp}(e^{j\omega})$ by one-half period (by π). Since the high frequencies in discrete-time are concentrated near π (and other odd multiples of π). So we get

$$H_{hp}(e^{j\omega}) = H_{lp}(e^{j(\omega-\pi)})$$

where $H_{hp}(e^{j\omega})$ is a highpass filter with cutoff frequency $\pi - \omega_c$.



$x(t)$ $x[2n]$

$x(t) \rightarrow x(at) \rightarrow X(j\omega)$

$a = 2$ อ้วน

$x[n] \rightarrow \cancel{x[an]}$ ไม่ได้ DTFT

$\cancel{a=2}$

$a = \frac{1}{2}$ $x[\frac{n}{2}]$ ขยาย

$x(\frac{1}{2}t)$

5. Differencing and Accumulation

$$x[n] \longleftrightarrow X(e^{j\omega})$$

From linearity properties

$$\text{DTFT}\{x[n] - x[n-1]\} = X(e^{j\omega}) - e^{-j\omega}X(e^{j\omega})$$

$$= (1 - e^{-j\omega})X(e^{j\omega})$$

Considering

$$y[n] = \sum_{m=-\infty}^{n} x[m]$$

$$y[n] - y[n-1] = x[n]$$

$$\boxed{y[n] \longleftrightarrow \frac{X(e^{j\omega})}{1-e^{-j\omega}} + \pi X(e^{j0}) \sum_{l=-\infty}^{\infty} \delta(\omega - 2\pi l)}$$

Impulse train on the right reflects dc or average value that can result from the summation

6. Time Expansion

From continuous-time, $x(at) \longleftrightarrow \frac{1}{|a|} X(\frac{j\omega}{a})$

For discrete-time, we cannot define $x[an]$ because

a might not be an integer.

$x[\frac{n}{k}]$

k — จ.น. เต็ม

★ Slow down the signal by choosing $a < 1$

★ if a is an integer other than ± 1, e.g., $x[2n]$

we do not speed up the signal. Since n can take on

only integer values, the signal $x[2n]$ consists of the

even samples of $x[n]$ alone.

$x[\frac{n}{3}]$ ✓

Let k be a positive integer, define the signal

$$x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n \text{ is a multiple of } k. \\ 0, & \text{if } n \text{ is not a multiple of } k. \end{cases}$$

$x_{(k)}[n]$ equals 0, unless n is a multiple of k, i.e., unless $n = rk$. Fourier transform of $x_{(k)}[n]$ is given by

$$X_{(k)}(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x_{(k)}[n] e^{-j\omega n} = \sum_{r=-\infty}^{\infty} x_{(k)}[rk]\, e^{-j\omega \boxed{rk}}$$

$x[n]$ $\quad \frac{n}{3}$

$x[3n]$

$x_{(3)}[n]$

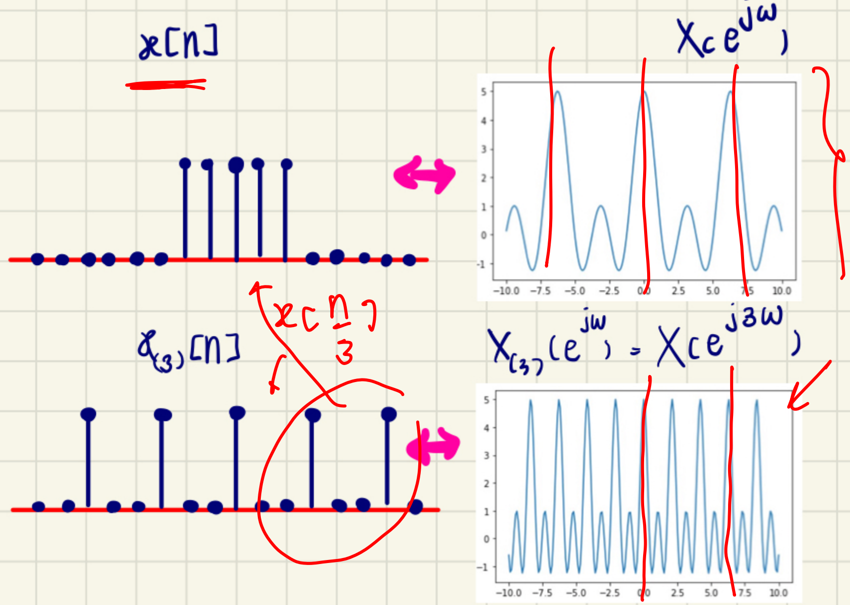
Since $x_{(k)}[rk] = x[r]$

$$X(e^{j\omega}) \qquad X_{(k)}(e^{j\omega}) = \sum_{r=-\infty}^{\infty} x[r] e^{-j(k\omega)r} = X(e^{jk\omega})$$

$$x[\tfrac{n}{3}] \longleftrightarrow \boxed{X(e^{j3\omega})}$$

$$x_{(k)}[n] \longleftrightarrow X(e^{jk\omega})$$

when $k>1$, the signal is spread out and slow down, its Fourier transform is compressed.
Since $X(e^{j\omega})$ is periodic with period 2π, $X(e^{jk\omega})$ is periodic with period $2\pi/k$.



7) Differentiation in Frequency

From $X(e^{jw}) = \sum_{n=-\infty}^{\infty} x[n] e^{-jwn}$

differentiate both sides, we obtain

$$\frac{d}{dw} X(e^{jw}) = \sum_{n=-\infty}^{\infty} x[n] (-jn) e^{-jwn}$$

$$-jn\, x[n] \longleftrightarrow \frac{d}{dw} X(e^{jw})$$

$$\boxed{n\, x[n] \longleftrightarrow j \frac{d}{dw} X(e^{jw})}$$

9) Convolution Property

$$y[n] = x[n] * h[n]$$

look at n

$$Y(e^{j\omega}) = \text{DTFT}\left\{ \sum_{k=-\infty}^{\infty} x[k]\, h[n-k] \right\}$$

$$= \sum_{k=-\infty}^{\infty} x[k] \underline{\sum_{n=-\infty}^{\infty} h[n-k]\, e^{-j\omega n}}$$

$$= \sum_{k=-\infty}^{\infty} x[k]\, \underline{H(e^{j\omega})}\, e^{-jk\omega}$$

$$= X(e^{j\omega})\, H(e^{j\omega})$$

$$\boxed{x[n] * h[n] \longleftrightarrow X(e^{j\omega})\, H(e^{j\omega})}$$

Example #1: Convolution: Given that

$$h[n] = \delta[n-n_0]$$

Frequency response is

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \delta[n-n_0]\, e^{-j\omega n}$$

$$= e^{-j\omega n_0}$$

For any input $x[n]$ with Fourier transform $X(e^{j\omega})$, Fourier transform of the output,

$$Y(e^{j\omega}) = e^{-j\omega n_0} X(e^{j\omega})$$

LTI system :

$h[n] = \alpha^n u[n] \qquad |\alpha| < 1$

$x[n] = \beta^n u[n] \qquad |\beta| < 1$

$x[n] \rightarrow \boxed{} \rightarrow y[n]$

① time domain

② freq → $y[n]$

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

$$= \left(\frac{1}{1-\alpha e^{-j\omega}}\right)\left(\frac{1}{1-\beta e^{-j\omega}}\right)$$

$\downarrow$

$y[n]$?

$h[n] \qquad H(e^{j\omega})$

$$\boxed{\alpha^n u[n] \leftrightarrow \frac{1}{1-\alpha e^{-j\omega}}}$$

$$= \frac{A}{1-\alpha e^{-j\omega}} + \frac{B}{1-\beta e^{-j\omega}} = \frac{1}{(1-\alpha e^{-j\omega})(1-\beta e^{-j\omega})}$$

หา A ① เอา $(1-\alpha e^{-j\omega}) \times$ สองข้าง

$$\frac{A\cancel{(1-\alpha e^{-j\omega})}}{\cancel{1-\alpha e^{-j\omega}}} + \frac{B\cancel{(1-\alpha e^{-j\omega})}}{1-\beta e^{-j\omega}} = \frac{1\cancel{(1-\alpha e^{-j\omega})}}{\cancel{(1-\alpha e^{-j\omega})}(1-\beta e^{-j\omega})}$$

② $e^{-j\omega} = \frac{1}{\alpha}$

$$A = \frac{1}{1-\beta\cdot\frac{1}{\alpha}} = \frac{\alpha}{\alpha-\beta}$$

$$B = -\frac{\beta}{\alpha-\beta}$$

$$Y(e^{j\omega}) = \frac{1}{\alpha-\beta}\left(\frac{\alpha}{1-\alpha e^{-j\omega}} - \frac{\beta}{1-\beta e^{-j\omega}}\right)$$

$$y[n] = \frac{1}{\alpha - \beta}\left(\alpha^{n+1} u[n] - \beta^{n+1} u[n]\right)$$

$\beta = \alpha$

$$\boxed{Y(e^{j\omega}) = \frac{1}{(1 - \alpha e^{-j\omega})^2}}$$

$y[n]$?

$$a^n u[n] \longleftrightarrow \frac{1}{1 - a e^{-j\omega}}$$

$$\boxed{(-jn)\,\alpha^n u[n]} \longleftrightarrow \frac{d}{d\omega}\left(\frac{1}{1 - \alpha e^{-j\omega}}\right)$$

$$\cancel{(-j)}\,n\,\alpha^n u[n] \longleftrightarrow \frac{1}{(1 - \alpha e^{-j\omega})^2} \quad (\cancel{-1})(\cancel{-}\alpha)(e^{-j\omega})(\cancel{-j})$$

$x[n+1]$

$\frac{1}{\alpha} x$

$$n\,\alpha^{n-1} u[n] \longleftrightarrow \frac{1}{(1 - \alpha e^{-j\omega})^2} \quad e^{-j\omega} \quad e^{j\omega}$$

$u[n]$

$$(n+1)\,\alpha^n u[n+1] \longleftrightarrow \frac{1}{(1 - \alpha e^{-j\omega})^2}$$

$-1 \quad 1$

$\neq$

$(n+1)\alpha^n u[n] \quad n = -1 \quad$ ได้ 0

cos

e^{jk}

k

a_k

cos

Ex #2 : Convolution

Consider an LTI system with impulse response

$$h[n] = \alpha^n u[n] \quad \text{with } |\alpha| < 1$$

Input to the system is

$$x[n] = \beta^n u[n] \quad \text{with } |\beta| < 1$$

Find $y[n]$

Solution

From $y[n] = x[n] * h[n]$

and $Y(e^{jw}) = X(e^{jw})\, H(e^{jw})$

we obtain $H(e^{jw}) = \dfrac{1}{1-\alpha e^{-jw}}$, and

$$X(e^{jw}) = \frac{1}{1-\beta e^{-jw}}$$

$$Y(e^{jw}) = \frac{1}{1-\alpha e^{-jw}} \cdot \frac{1}{1-\beta e^{-jw}}$$

$$Y(e^{jw}) = \frac{A}{1-\alpha e^{-jw}} + \frac{B}{1-\beta e^{-jw}} \quad \text{—} \ast\ast$$

Using partial fraction expansion to find A and B

To find A, multiply 2 sides by $(1-\alpha e^{-jw})$ and replace $e^{-jw} = \frac{1}{\alpha}$ to remove variable B,

$$\frac{\cancel{1-\alpha e^{-jw}}}{\cancel{(1-\alpha e^{-jw})}(1-\beta e^{-jw})} = A + \left(\frac{1-\alpha e^{-jw}}{1-\beta e^{-jw}}\right) B$$

$$\frac{1}{1-\beta/\alpha} = A + 0 \cdot B$$

$$A = \frac{\alpha}{\alpha - \beta}$$

Similarly, to find B, multiply ** by $1-\beta e^{-jw}$ and replace $e^{-jw} = \frac{1}{\alpha}$ to remove A

$$B = \frac{\beta}{\beta - \alpha} = -\frac{\beta}{\alpha - \beta}$$

#3

From linearity property, we get

$$y[n] = \frac{\alpha}{\alpha-\beta}\,\alpha^{n}u[n] + \overset{\frac{-\beta}{\alpha-\beta}}{\frac{\beta}{\beta-\alpha}}\,\beta^{n}u[n]$$

$$= \frac{1}{\alpha-\beta}\left(\alpha^{n+1}u[n] - \beta^{n+1}u[n]\right)$$

Example #2 part II, Find y[n] when $\alpha = \beta$

Solution #2

$$Y(e^{j\omega}) = \left(\frac{1}{1-\alpha e^{-j\omega}}\right)^{2}$$

$$= (1-\alpha e^{-j\omega})^{-2}$$

From $\mathcal{F}\{\alpha^{n}u[n]\} = \frac{1}{1-\alpha e^{-j\omega}}$

$= \left(1-\alpha e^{-j\omega}\right)^{-1}$

$$\alpha^{n}u[n] \leftrightarrow (1-\alpha e^{-j\omega})^{-1}$$

$$\frac{d}{d\omega}(1-\alpha e^{-j\omega})^{-1}$$

$$= (1-\alpha e^{-j\omega})^{-2}\cdot(-\alpha)(-j)(e^{-j\omega})(-1)$$

$$\therefore (1-e^{-j\omega})^{-2} = \frac{j}{\alpha}e^{j\omega}\frac{d}{d\omega}(1-e^{-j\omega})^{-1}$$

จาก $\alpha^{n}u[n] \leftrightarrow \frac{1}{1-e^{-j\omega}}$

Differentiation:

$$n\alpha^{n}u[n] \leftrightarrow j\frac{d}{d\omega}\frac{1}{1-e^{-j\omega}}$$

time-shifting

$$(n+1)\alpha^{n+1}u[n+1] \leftrightarrow j\,e^{j\omega}\frac{d}{d\omega}(1-e^{-j\omega})^{-1}$$

$$(n+1)\alpha^{n}u[n+1] \leftrightarrow \frac{j}{\alpha}e^{j\omega}\frac{d}{d\omega}(1-e^{-j\omega})^{-1}$$

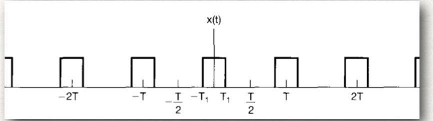
y[n] is still zero prior to n = 0, so we can use u[n]

$$\boxed{y[n] = (n+1)\alpha^{n}u[n]}$$

#4

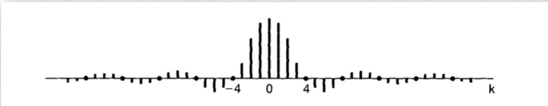
COMPARISON

FOURIER SERIES

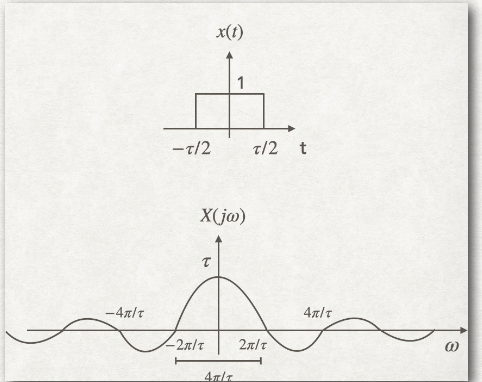


$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t}$$

$$a_k = \frac{1}{T}\int_{<T>} x(t)e^{-jk\omega_0 t}\,dt = \frac{1}{T}\int_0^T x(t)e^{-jk(2\pi/T)t}dt$$



FOURIER TRANSFORM

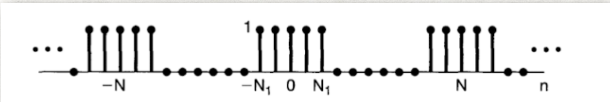


$$x(t) = \frac{1}{2\pi}\int_{\omega=-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega$$

$$X(j\omega) = \int_{t=-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$$

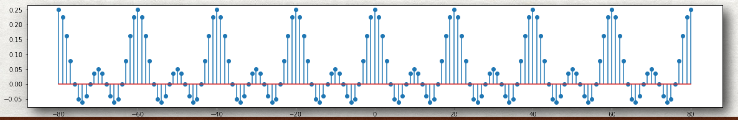
CONTINUOUS-TIME

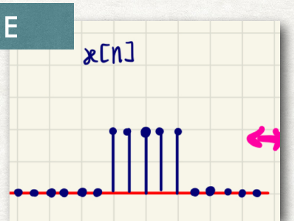
DISCRETE-TIME



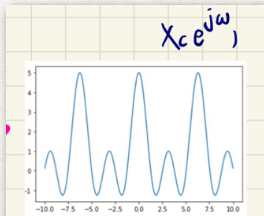
$$x[n] = \sum_{k=<N>} a_k e^{jk(\omega_0)n} = \sum_{k=<N>} a_k e^{jk(2\pi/N)n}$$

$$a_k = \frac{1}{N}\sum_{k=<N>} x[n]e^{-jk\omega_0 n} = \frac{1}{N}\sum_{k=<N>} x[n]e^{-jk(2\pi/N)n}$$





$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}\,d\omega$$



$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

TABLE 5.1 PROPERTIES OF THE DISCRETE-TIME FOURIER TRANSFORM

Section	Property	Aperiodic Signal	Fourier Transform
		$x[n]$ $y[n]$	$\left.\begin{array}{l}X(e^{j\omega})\\Y(e^{j\omega})\end{array}\right\}$ periodic with period 2π
5.3.2	Linearity	$ax[n]+by[n]$	$aX(e^{j\omega})+bY(e^{j\omega})$
5.3.3	Time Shifting	$x[n-n_0]$	$e^{-j\omega n_0}X(e^{j\omega})$
5.3.3	Frequency Shifting	$e^{j\omega_0 n}x[n]$	$X(e^{j(\omega-\omega_0)})$
5.3.4	Conjugation	$x^*[n]$	$X^*(e^{-j\omega})$
5.3.6	Time Reversal	$x[-n]$	$X(e^{-j\omega})$
5.3.7	Time Expansion	$x_{(k)}[n]=\begin{cases}x[n/k], & \text{if } n=\text{multiple of } k\\ 0, & \text{if } n\neq\text{multiple of } k\end{cases}$	$X(e^{jk\omega})$
5.4	Convolution	$x[n]*y[n]$	$X(e^{j\omega})Y(e^{j\omega})$
5.5	Multiplication	$x[n]y[n]$	$\frac{1}{2\pi}\int_{2\pi}X(e^{j\theta})Y(e^{j(\omega-\theta)})d\theta$
5.3.5	Differencing in Time	$x[n]-x[n-1]$	$(1-e^{-j\omega})X(e^{j\omega})$
5.3.5	Accumulation	$\sum_{k=-\infty}^{n}x[k]$	$\frac{1}{1-e^{-j\omega}}X(e^{j\omega})$ $+\pi X(e^{j0})\sum_{k=-\infty}^{+\infty}\delta(\omega-2\pi k)$
5.3.8	Differentiation in Frequency	$nx[n]$	$j\frac{dX(e^{j\omega})}{d\omega}$
5.3.4	Conjugate Symmetry for Real Signals	$x[n]$ real	$\begin{cases}X(e^{j\omega})=X^*(e^{-j\omega})\\ \mathcal{R}e\{X(e^{j\omega})\}=\mathcal{R}e\{X(e^{-j\omega})\}\\ \mathcal{I}m\{X(e^{j\omega})\}=-\mathcal{I}m\{X(e^{-j\omega})\}\\ \lvert X(e^{j\omega})\rvert=\lvert X(e^{-j\omega})\rvert\\ \sphericalangle X(e^{j\omega})=-\sphericalangle X(e^{-j\omega})\end{cases}$
5.3.4	Symmetry for Real, Even Signals	$x[n]$ real an even	$X(e^{j\omega})$ real and even
5.3.4	Symmetry for Real, Odd Signals	$x[n]$ real and odd	$X(e^{j\omega})$ purely imaginary and odd
5.3.4	Even-odd Decomposition of Real Signals	$x_e[n]=\mathcal{E}v\{x[n]\}$ $[x[n]$ real$]$ $x_o[n]=\mathcal{O}d\{x[n]\}$ $[x[n]$ real$]$	$\mathcal{R}e\{X(e^{j\omega})\}$ $j\mathcal{I}m\{X(e^{j\omega})\}$
5.3.9	Parseval's Relation for Aperiodic Signals		

$$\sum_{n=-\infty}^{+\infty}|x[n]|^2=\frac{1}{2\pi}\int_{2\pi}|X(e^{j\omega})|^2d\omega$$

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TABLE 5.2 BASIC DISCRETE-TIME FOURIER TRANSFORM PAIRS

Signal	Fourier Transform	Fourier Series Coefficients (if periodic)
$\sum_{k=\langle N\rangle} a_k e^{jk(2\pi/N)n}$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta\left(\omega - \frac{2\pi k}{N}\right)$	a_k
$e^{j\omega_0 n}$	$2\pi \sum_{l=-\infty}^{+\infty} \delta(\omega - \omega_0 - 2\pi l)$	(a) $\omega_0 = \frac{2\pi m}{N}$ $a_k = \begin{cases} 1, & k = m, m \pm N, m \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}$ (b) $\frac{\omega_0}{2\pi}$ irrational $\Rightarrow$ The signal is aperiodic
$\cos \omega_0 n$	$\pi \sum_{l=-\infty}^{+\infty} \{\delta(\omega - \omega_0 - 2\pi l) + \delta(\omega + \omega_0 - 2\pi l)\}$	(a) $\omega_0 = \frac{2\pi m}{N}$ $a_k = \begin{cases} \frac{1}{2}, & k = \pm m, \pm m \pm N, \pm m \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}$ (b) $\frac{\omega_0}{2\pi}$ irrational $\Rightarrow$ The signal is aperiodic
$\sin \omega_0 n$	$\frac{\pi}{j} \sum_{l=-\infty}^{+\infty} \{\delta(\omega - \omega_0 - 2\pi l) - \delta(\omega + \omega_0 - 2\pi l)\}$	(a) $\omega_0 = \frac{2\pi r}{N}$ $a_k = \begin{cases} \frac{1}{2j}, & k = r, r \pm N, r \pm 2N, \ldots \\ -\frac{1}{2j}, & k = -r, -r \pm N, -r \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}$ (b) $\frac{\omega_0}{2\pi}$ irrational $\Rightarrow$ The signal is aperiodic
$x[n] = 1$	$2\pi \sum_{l=-\infty}^{+\infty} \delta(\omega - 2\pi l)$	$a_k = \begin{cases} 1, & k = 0, \pm N, \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}$
Periodic square wave $x[n] = \begin{cases} 1, & \lvert n\rvert \le N_1 \\ 0, & N_1 < \lvert n\rvert \le N/2 \end{cases}$ and $x[n+N] = x[n]$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta\left(\omega - \frac{2\pi k}{N}\right)$	$a_k = \frac{\sin[(2\pi k/N)(N_1 + \frac{1}{2})]}{N \sin[2\pi k/2N]}, \ k \ne 0, \pm N, \pm 2N, \ldots$ $a_k = \frac{2N_1 + 1}{N}, \ k = 0, \pm N, \pm 2N, \ldots$
$\sum_{k=-\infty}^{+\infty} \delta[n - kN]$	$\frac{2\pi}{N} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{N}\right)$ $\underline{X(e^{j\omega})}$	$a_k = \frac{1}{N}$ for all k
$a^n u[n], \quad \lvert a\rvert < 1$	$\frac{1}{1 - ae^{-j\omega}}$	—
$x[n] \begin{cases} 1, & \lvert n\rvert \le N_1 \\ 0, & \lvert n\rvert > N_1 \end{cases}$	$\frac{\sin[\omega(N_1 + \frac{1}{2})]}{\sin(\omega/2)}$	—
$\frac{\sin Wn}{\pi n} = \frac{W}{\pi} \operatorname{sinc}\left(\frac{Wn}{\pi}\right)$ $0 < W < \pi$	$X(\omega) = \begin{cases} 1, & 0 \le \lvert\omega\rvert \le W \\ 0, & W < \lvert\omega\rvert \le \pi \end{cases}$ $X(\omega)$ periodic with period 2π	—
$\delta[n]$	1	—
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{+\infty} \pi\delta(\omega - 2\pi k)$	—
$\delta[n - n_0]$	$e^{-j\omega n_0}$	—
$(n+1)a^n u[n], \quad \lvert a\rvert < 1$	$\frac{1}{(1 - ae^{-j\omega})^2}$	—
$\frac{(n+r-1)!}{n!(r-1)!} a^n u[n], \quad \lvert a\rvert < 1$	$\frac{1}{(1 - ae^{-j\omega})^r}$	—

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$k \cdot w_0$

$\cos \frac{2\pi}{6} n \rightarrow \frac{e^{j\frac{2\pi}{6}n}}{2} + \frac{e^{-j\frac{2\pi}{6}n}}{2}$

$\frac{1}{2}$ a_1 a_{-1}

REFERENCES

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